GUILLAUME RODIER

MASc Candidate — Aerospace Engineering

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CONTACT

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TECHNICAL SKILLS

Programming
- C++ / Python
- MATLAB / Simulink
- Bash / Linux
- Arduino

Engineering & Simulation
- CFD — LES / RANS
- Tabulated chemistry (FPI)
- SolidWorks / CAD
- 3D Printing / Fabrication

Embedded Systems
- NVIDIA Jetson Nano
- Embedded Linux

Standards & Quality
- ISO 13485 / QMS

LANGUAGES

- French — Native
- English — Fluent

CERTIFICATIONS

- National Lifeguard License
- CPR-AED Level C (18 h)
- Anaphylaxis Care (1.5 h)
- PADI Open Water Diver
- Avalanche Safety — Level 1

SPORT

Triathlon
- U of T & McGill Tri Club
- 2026 World Championship Qualifier
  Pontevedra, Spain · Olympic dist.
- ~11–14 hrs / week

Ice Hockey
- St-Lambert Red-Wings
- 10 seasons · Competitive A/B

Badminton
- Ste-Julie Club · 2012–2020
- Regional tournaments

EDUCATION

MASc in Aerospace Engineering — *Sep 2020 – Sep 2026*
University of Toronto · CFD & Propulsion Group · Supervised by Prof. Clinton Groth
- Specialization in aerospace propulsion with a focus on hydrogen combustion for aviation gas turbine engines.
- Simulating premixed hydrogen flames via LES/RANS coupled with tabulated chemistry (FPI) to study turbulent flame behavior.
- Implementing a C++/Python correction within a Navier-Stokes PCM-FPI solver to account for hydrogen's preferential diffusion, a critical physical effect in lean H_2 combustion.

BEng. in Bioengineering — *Sep 2020 – Sep 2024*
McGill University · Specialization in Biological Information & Computation
- Graduated with distinction

RELEVANT WORK EXPERIENCE

Product Engineering Intern — *May 2023 – Sep 2023*
Rogue Research
- Generated, edited, and maintained QMS documentation including procedures, work instructions, and incoming quality controls for Transcranial Magnetic Stimulation devices.
- High attention to regulatory detail in full alignment with ISO 13485 design control standards, supporting the company's medical device certification process.

Mechanical Contractor — *April 2022 – Oct 2022*
SAJB Industrial Automation
- Executed industrial automation contracts for CHEP Canada and O-I Glass: installed trapped-key interlock systems to secure access to production machinery across multiple Canadian facilities.
- Deployed a robotic production line for box-insert formation with Fanuc robots and conveyor systems, commissioned on-site with full functional testing.

DESIGN PROJECTS

Zimmer Biomet — Capstone Design Project — *Sep 2023 – May 2024*
Four-student interdisciplinary team
- Validated sensor accuracy for robot-assisted total knee arthroplasty (TKA), a procedure requiring sub-millimeter positioning precision.
- Designed and built a validation jig using optical infrared tracker constellations to confirm the robotic knee cutting-guide’s accuracy.
- Produced full technical documentation, test protocols, and a validation report.

McGill Robotics Team — Mars Rover Challenge — *Dec 2021 – Sep 2022*
McGill Robotics · Mechanical Subsystems
- Designed a compact electrical enclosure resolving high-density thermal management challenges for multiple stacked PCBs operating in a sealed rover chassis.
- Developed a servo-actuated pan-tilt camera system providing 360° peripheral vision for navigation through complex simulated Martian terrain.

Solid Propellant Model Rocket — IB Project — *Sep 2017 – Jun 2018*
- Designed and built a solid-propellant rocket powered by a KNO_3-based propellant, including the nozzle, combustion chamber, and recovery system.
- Modelled the de Laval nozzle and parachute ejection mechanism in SolidWorks; manufactured components using lathe, milling machine, and soldering equipment.
- Integrated GPS tracking and an automated parachute deployment system via Arduino. Achieved 2nd place at the regional competition.

AWARDS & DISTINCTIONS

NSERC Undergraduate Student Research Award (USRA) — *Jun 2024*
- Federal research grant to investigate virus propagation using fluid dynamics principles within Dr. Caroline Wagner's lab, addressing airborne respiratory disease transmission.

KEY COURSES

Aerospace
- Turbulence Modelling
- Combustion
- Finite Volume Methods
- Advanced Flight Dynamics
- Computational Fluid Dynamics (audited)
- Aerospace Propulsion (audited)

Bioengineering
- Machine Learning
- Medical Imaging
- Signals & Systems
- Structure of materials
- Fluid Mechanics I & II
- Analytical mechanics
- Computer Programming for Physical Sciences & Engineering

TOOLS & SOFTWARE

- Tecplot 360
- Git / GitHub
- LaTeX
- Arduino IDE

AWARDS

NSERC USRA
Federal research grant

Ross Ritchie Scholarship
Faculty of Engineering, McGill

PROFILE

Propulsion · CFD · LES/RANS
H_2 Combustion · Navier-Stokes
Embedded Systems · C++ · Python
SolidWorks · 3D Printing

Ross Ritchie Family Scholarship *Jul 2022*
- Exclusive award from the Faculty of Engineering and Department of Athletics at McGill for outstanding academic performance and involvement with the St-Lambert Hockey Team.

RESEARCH EXPERIENCE

Student Researcher — Ophthalmology *Oct 2023 – Sep 2024*
Montreal Neurological Institute · Supervised by Dr. Leonard Levin
- Developed a standalone pupillometer in C++ on embedded Linux to detect relative afferent pupillary defects (RAPD) and support neurological diagnosis in environments with limited medical access, including combat zones.
- Device equipped with infrared illumination and Basler cameras, programmed on an NVIDIA Jetson Nano microcontroller for real-time image acquisition and processing.
- Designed the full enclosure in SolidWorks, optimized for 3D printing and field deployment.

Summer Undergraduate Student Researcher — Fluid Dynamics *May 2024 – Sep 2024*
McGill University · Supervised by Prof. Caroline Wagner
- Improved the design of a temperature and humidity-controlled chamber to study mucus droplet propagation under high-speed imaging.

Conceived and integrated an automated droplet ejection system to study the influence of environmental conditions and viruses on droplet size, contributing to understanding seasonal respiratory virus peaks.

SOFTWARE PROJECTS

Protein Secondary Structure Prediction *Fall 2023*
McGill University · Bioinformatics
- Python algorithm to predict secondary protein structure (α-helices, β-sheets) from primary amino acid sequences, drawing from Naïve Bayes, Chou-Fasman, and GOR algorithms.

Neural Network for Medical Image Analysis *Winter 2023*
McGill University · Computational Biology
- K-Means unsupervised segmentation of cell types in medical microscopy image stacks in Python, followed by HoG feature extraction.
- Implemented an SVM classifier to identify cancerous cells within the segmented cell population.

VOLUNTEERING

- **Ontario Athletics —** Utilized high-speed cameras to time track and field races, giving back to the competitive athletics community.
- **Tutoring —** Tutored CEGEP students in Differential Calculus and Organic Chemistry, helping them navigate challenging coursework.
- **CHUM Hospital —** Led group activities (10+ patients) across toxicology and psychiatry units — games, conversations, and rooftop gatherings to foster a supportive environment.
- **School of the Greats —** Mentored 15+ elementary students from disadvantaged backgrounds through homework support and hands-on science lab activities.

SPORT

Triathlon — University of Toronto & McGill Tri Club *2023 – 2026*
- Qualified for the 2026 World Triathlon Championship (Olympic distance) in Pontevedra, Spain.
- Maintains year-round structured training: 8 sessions per week across swim, bike, and run (~11–14 hours/week).

St-Lambert Red-Wings — Competitive Ice Hockey *2014 – 2023*
- Played competitive A/B level hockey for 10 consecutive seasons, including weekly practices, games, tournaments, and provincial playoff series.

Ste-Julie Badminton Club *2012 – 2020*
- Trained 6 sessions per week in singles and doubles formats; competed in regional-level tournaments throughout Quebec.

SKILLS & HOBBIES

Classical Guitar *10 Years*
- Ten years of classical guitar study, gaining proficiency in sheet music reading and fingerpicking, providing a creative pursuit beyond science.

Backcountry Skiing *5 Years*
- Skied challenging alpine terrain across Canada, from Quebec to the Rockies, including Canmore, Banff National Park, Kananaskis Country, Whistler, and North Vancouver.